# Number Systems in C++

# Lab 4

# Objective

- In this lab, we will put some of the concepts discussed in Chapters 1–3 into practical use by writing a C++ program that works with different number systems.
- What We Will Practice:
  - Binary, Octal, Decimal, and Hexadecimal number systems
  - Converting values between different number systems
  - Variables and data types
  - Input and output using cin and cout
  - Strings
  - `switch` statements (part of chapter 4 but partially introducing it to you)
  - Using C++ libraries
  - Using built-in C++ functions and manipulators

# Number Systems Review

- Computers can represent numbers using different bases.

| Number System | Base | Digits |
|---|---|---|
| Binary | 2 | 0–1 |
| Octal | 8 | 0–7 |
| Decimal | 10 | 0–9 |
| Hexadecimal | 16 | 0–9, A–F |

- Example:
  - Binary: 11111111
  - Octal: 377
  - Decimal: 255
  - Hexadecimal: FF
- All four represent the same numerical value.

# Task

- Write a C++ program that allows the user to:
- Select the number system of the input:
  - Binary
  - Octal
  - Decimal
  - Hexadecimal
- Enter a number in the selected number system.
- Convert the number and display its equivalent in:
  - Binary
  - Octal
  - Decimal
  - Hexadecimal

# Example:

Choose the number system of your input:

1. Binary
2. Octal
3. Decimal
4. Hexadecimal

Enter your choice: `1`

Enter the number: `11111111`

Binary: `11111111`

Octal: `377`

Decimal: `255`

Hexadecimal: `ff`

# Step 1: Required Libraries

```
#include <iostream>     //<iostream>  → cin and cout

#include <string>      // <string>    → string
#include <bitset>      //<bitset>    → display a number in
binary

using namespace std;
```

# Step 2: Declare Our Variables

```
int main()
{
    string input;
    int choice;
    int number;
```

input → stores what the user enters

choice → stores the user's menu choice

number → stores the converted numerical value

**Question:** Why are we storing input as a string rather than immediately storing it as an int?

Because FF cannot be entered directly into a normal decimal int using cin.

# Step 3: Display the Menu

```
cout << "Choose the number system of your input:" << endl;
cout << "1. Binary" << endl;
cout << "2. Octal" << endl;
cout << "3. Decimal" << endl;
cout << "4. Hexadecimal" << endl;

cout << "Enter your choice: ";
cin >> choice;

cout << "Enter the number: ";
cin >> input;
```

# Step 4: Use a switch

- Now we determine which number system the user selected.

```
switch (choice)
{
    case 1:
        number = stoi(input, nullptr, 2);
        break;

    case 2:
        number = stoi(input, nullptr, 8);
        break;

    case 3:
        number = stoi(input, nullptr, 10);
        break;

    case 4:
        number = stoi(input, nullptr, 16);
        break;

    default:
        cout << "Invalid choice." << endl;
        return 0;
}
```

# Understanding `stoi()`

- Consider:
  `number = stoi(input, nullptr, 2);`

`stoi` means: String to Integer. The final argument tells C++ which base the input uses.
For example

- 2 → Binary
- 8 → Octal
- 10 → Decimal
- 16 → Hexadecimal

```
stoi("1010", nullptr, 2)   // produces 10 in decimal
stoi(string, position, base)
```

`Position = nullptr` means we are not pointing to any address
`Position = &position` means we are pointing to an address
Example:

Number: 123
Stopped at position: 3
It stops when it reaches ‘a’

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
    string input = "123abc";
    size_t position;

    int number = stoi(input, &position, 10);

    cout << "Number: " << number << endl;
    cout << "Stopped at position: " << position << endl;

    return 0;
}
```

# Step 5: Display the Conversions

- Once the input has been converted to number, we can display it in different number systems.

```
cout << "Binary:      " << bitset<8>(number) << endl;
cout << "Octal:       " << oct << number << endl;
cout << "Decimal:     " << dec << number << endl;
cout << "Hexadecimal: " << hex << number << endl;
```

- The above code displays:
  - bitset<8> → displays 8 binary bits
  - oct → displays in base 8
  - dec → displays in base 10
  - hex → displays in base 16

# Complete Program

```
#include <iostream>
#include <string>
#include <bitset>
using namespace std;

int main()
{
    string input;
    int choice;
    int number;

    cout << "Choose the number system of your input:" << endl;
    cout << "1. Binary" << endl;
    cout << "2. Octal" << endl;
    cout << "3. Decimal" << endl;
    cout << "4. Hexadecimal" << endl;

    cout << "Enter your choice: ";
    cin >> choice;

    cout << "Enter the number: ";
    cin >> input;

    switch (choice)
    {
        case 1:
            number = stoi(input, nullptr, 2);
            break;

        case 2:
            number = stoi(input, nullptr, 8);
            break;

        case 3:
            number = stoi(input, nullptr, 10);
            break;

        case 4:
            number = stoi(input, nullptr, 16);
            break;

        default:
            cout << "Invalid choice." << endl;
            return 0;
    }

    cout << endl;
    cout << "Binary:      " << bitset<8>(number) << endl;
    cout << "Octal:       " << oct << number << endl;
    cout << "Decimal:     " << dec << number << endl;
    cout << "Hexadecimal: " << hex << number << endl;

    return 0;
}
```

# Test the Program

- Test 1
  - Choice: 1
  - Input: 1010
- Expected:
  - Binary: 00001010
  - Octal: 12
  - Decimal: 10
  - Hexadecimal: a

- Test 2
  - Choice: 4
  - Input: FF
- Expected:
  - Binary: 11111111
  - Octal: 377
  - Decimal: 255
  - Hexadecimal: ff

# What Did We Use?

Number Systems

+

Variables and Data Types

+

Input and Output

+

Strings

+

Libraries

+

switch / case / break

+

Built-in Functions

# In-Class Exercise – Number Systems

Task: Write a C++ program that performs the following:

1. Ask the user to enter three integers. Example Output
2. Calculate and display the sum of the three numbers.
3. Calculate the average of the three numbers.
   - Use `static_cast<double>()` to avoid integer division.
   - Display the average to 2 decimal places using fixed and setprecision().
4. Ask the user how they would like to display the sum:
   1. Binary
   2. Octal
   3. Hexadecimal
5. Use a switch statement to display the sum in the selected number system.
6. Display "Invalid choice" if the user enters a choice other than 1, 2, or 3.

```
Example Output:
Enter three integers: 10 15 20

Sum: 45
Average: 15.00

Choose how to display the sum:
1. Binary
2. Octal
3. Hexadecimal

Enter your choice: 2

Sum in Octal: 55
```

# Solution

```
#include <iostream>
#include <iomanip>
#include <bitset>
using namespace std;

int main()
{
    int num1, num2, num3;
    int sum;
    double average;
    int choice;

    cout << "Enter three integers: ";
    cin >> num1 >> num2 >> num3;

    sum = num1 + num2 + num3;

    average = static_cast<double>(sum) / 3;

    cout << fixed << setprecision(2);
    cout << "Average: " << average << endl;
    cout << "Sum: " << sum << endl;

    cout << endl;
    cout << "Choose how to display the sum:" << endl;
    cout << "1. Binary" << endl;
    cout << "2. Octal" << endl;
    cout << "3. Hexadecimal" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice)
    {
        case 1:
            cout << "Sum in Binary: "
                 << bitset<8>(sum) << endl;
            break;

        case 2:
            cout << "Sum in Octal: "
                 << oct << sum << endl;
            break;

        case 3:
            cout << "Sum in Hexadecimal: "
                 << hex << sum << endl;
            break;

        default:
            cout << "Invalid choice." << endl;
    }

    return 0;
}
```

If you enter: 10 15 20

sum = 45
average = 15.00

If you choose 2:
Average: 15.00
Sum: 45

Choose how to display the sum:
1. Binary
2. Octal
3. Hexadecimal

Enter your choice: 2
Sum in Octal: 55